



**INDUSTRIAL
SUMMIT 2023**

SHENZHEN, CHINA | 28 SEPTEMBER
POWERING YOUR SUSTAINABLE INNOVATION



基于AI的智能电弧故障 分断器(AFCI)

潘瑞锋 & 李婷

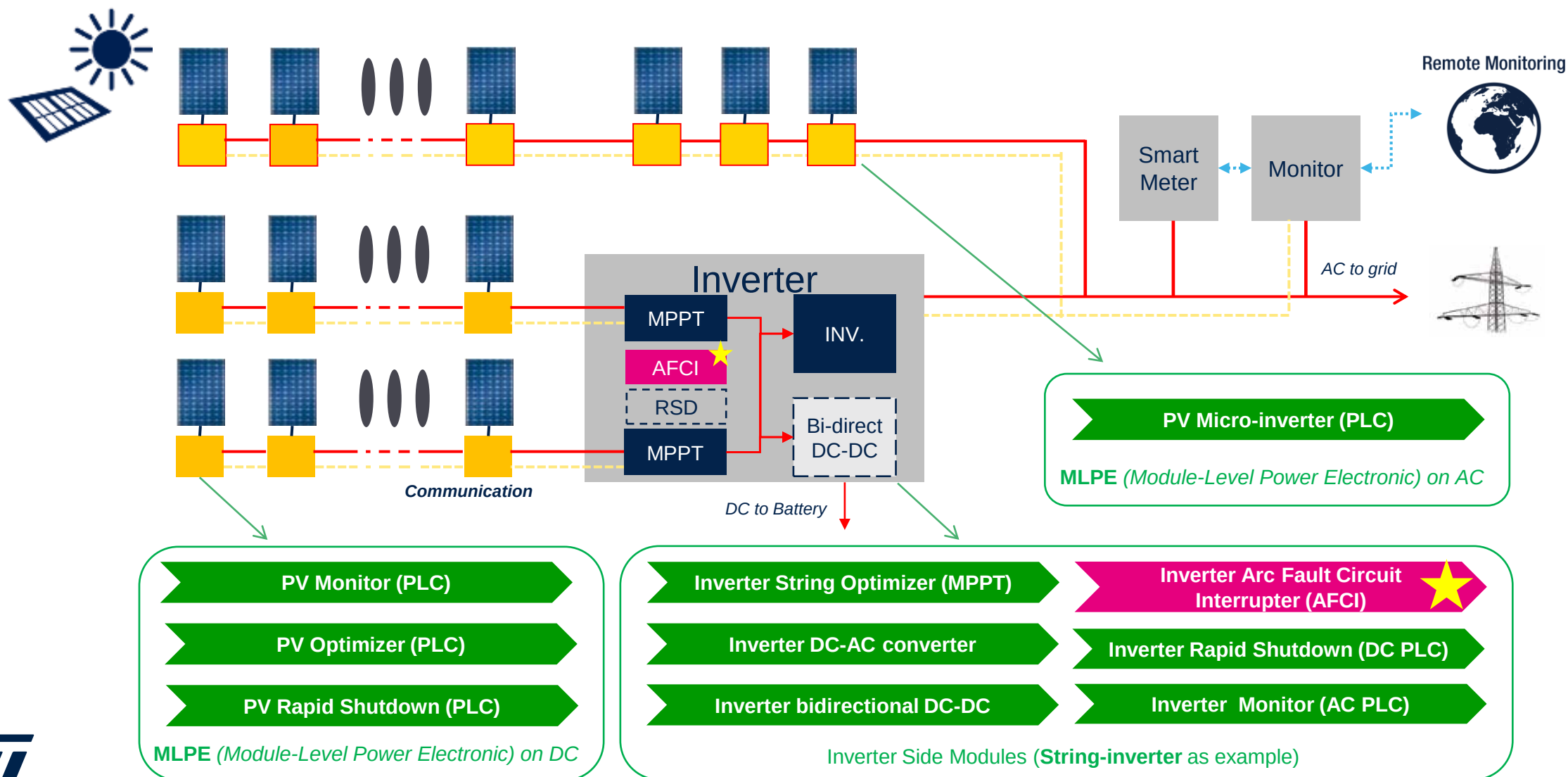
Power & Energy
Competence
Center



AI
Competence
Center



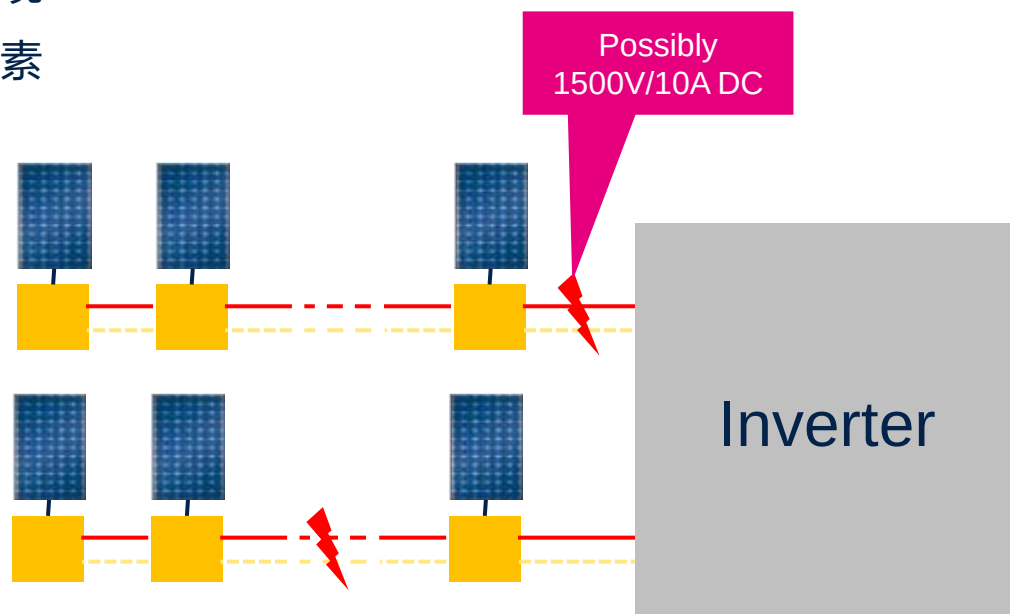
光伏系统中ST的参考设计



电弧故障的危害

电弧产生的诱因

- 设备老化
- 多样天气
- 恶略环境
- 人为因素
-



电弧故障检测标准- UL1699B

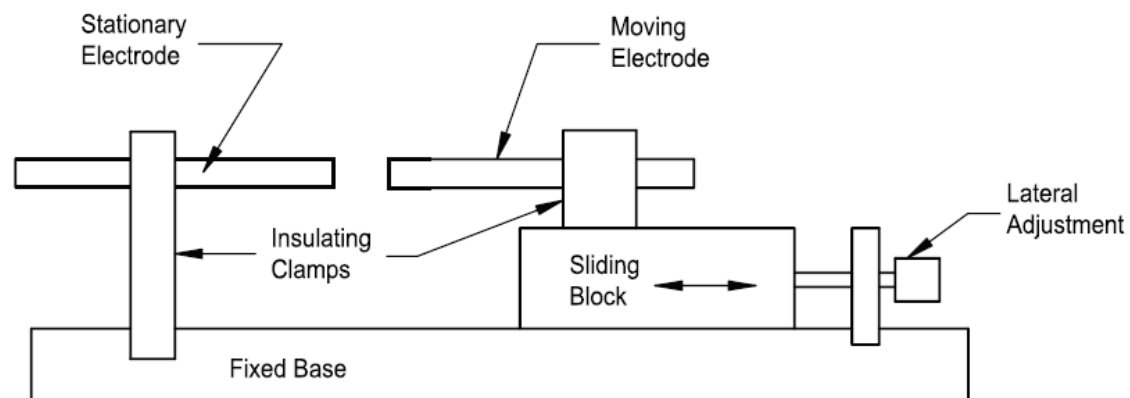


安全标准

- 光伏 (PV) 直流电弧故障电路保护

Figure 29.10

Arc generator



Arcing test conditions

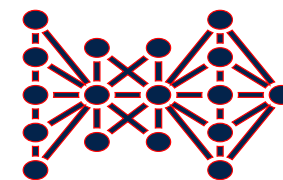
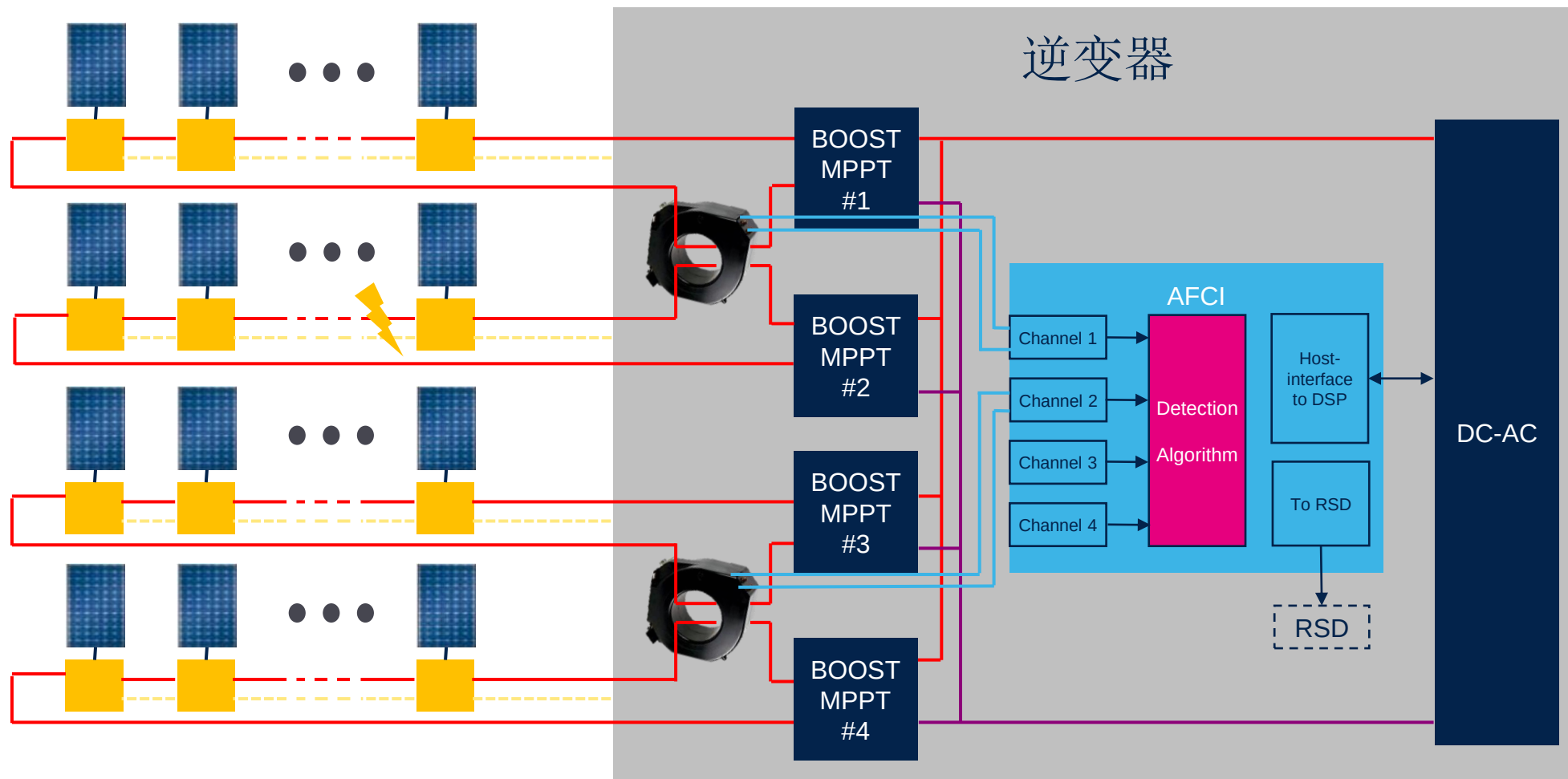
Test no.	Minimum I_{arc} (A) ^a	I_{mpp} (A)	Sep. rate (mm/s)	V_{mpp} (V) ^b	V_{oc} (V) ^b	R_{tot} (ohms) ^b	Gap (mm)
1	2.5	3.0	2.5	312.0	480.0	56.0	0.8
2	7.0	8.0	5.0	318.0	490.0	21.0	0.8
3 ^c	14.0	16.0	5.0	318.0	490.0	11.0	1.1
4	7.0	8.5	5.0	607.0	810.0	24.0	2.5
Single module							
5	2.5	3.0	2.5	31.2	48.0	5.6	0.8
6	7.0	8.0	5.0	31.8	49.0	2.1	0.8

^a I_{arc} values in Table 29.2 are representative of realistic arc events with one or two strings at full and low irradiance. It is expected that the AFCI shall meet compliance criteria below at intermediate current levels also.

^b These values are approximate.

^c Required for arcs in series with two parallel strings.

AFCI在光伏系统中的应用



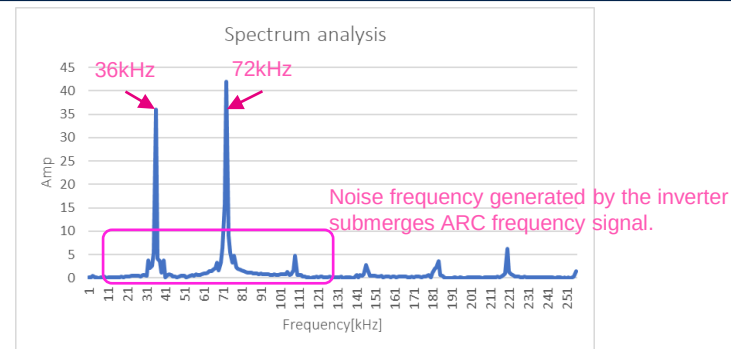
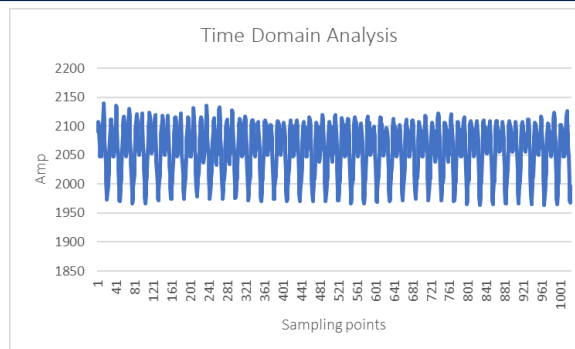
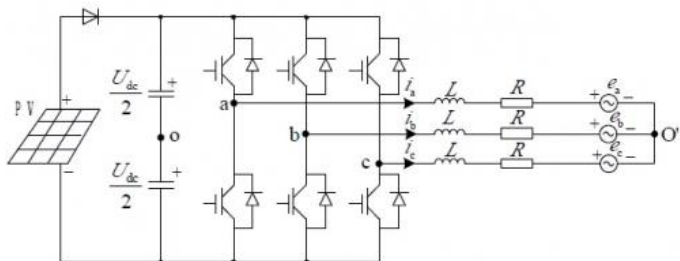
神经网络

STM32
Cube.AI

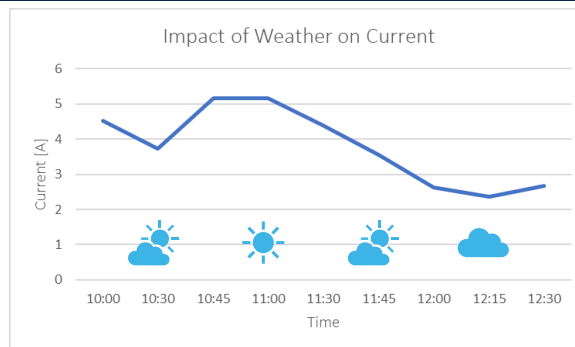
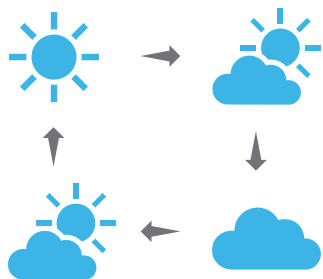
基于STM32+AI的电弧故障分断器是经过现场验证且面向未来的产品方案。

AFCI系统设计难点

逆变器噪声

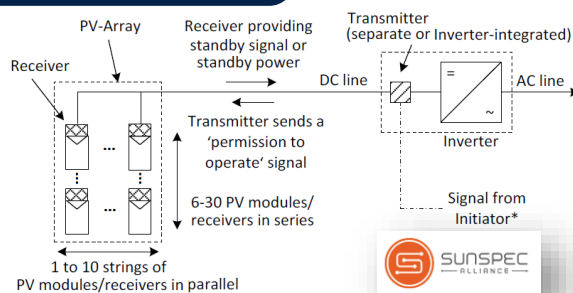


天气变化



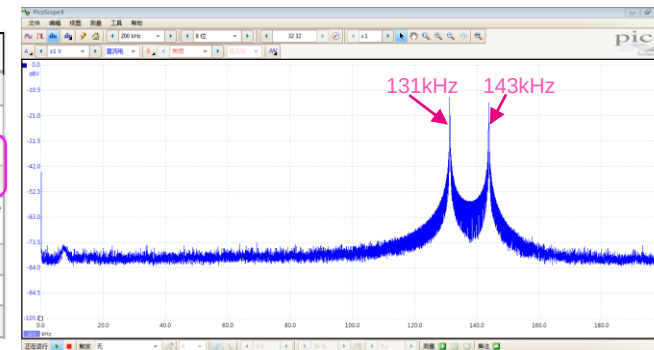
Date	Time	Weather	Voltage [V]	Current [A]	Power [KW]
15/07/2023	10.00	Nice, little cloudy	315	4.52	1.42
	10.30	Nice, little cloudy	277	3.74	1.04
	10.45	Nice, little cloudy	301	5.16	1.56
	11.00	Nice, little cloudy	300	5.16	1.55
	11.30	Nice, little cloudy	300	4.39	1.26
	11.45	Nice, little cloudy	320	3.55	1.10
	12.00	Cloudy	317	2.62	0.83
	12.15	Cloudy	334	2.36	0.79
	12.30	Cloudy	331	2.67	0.87

快速关断

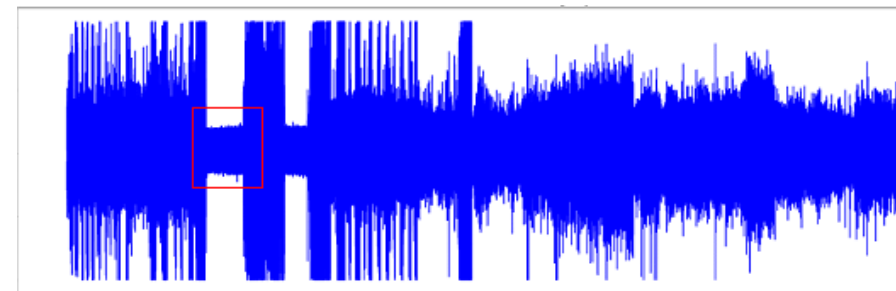
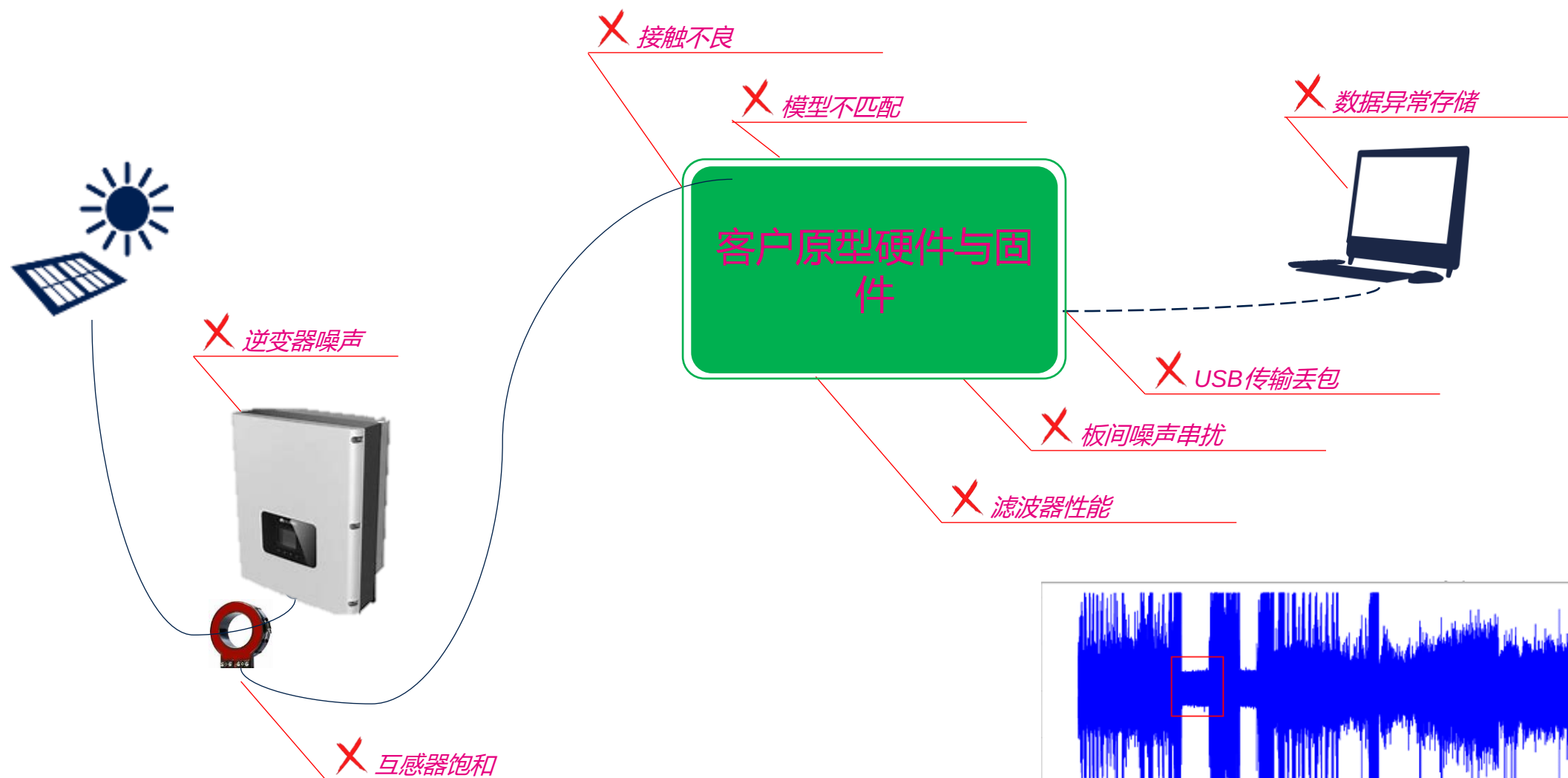


Requirements of the Power Line Communication

Symbol	Transmitter Specification	Min.	Nom.	Max.	Unit	Remark
W_1	Logic 1 Code Word	{-1, -1, -1, +1, +1, -1, +1, -1, +1}				+1 = mark, -1 = space
W_0	Logic 0 Code Word	{+1, +1, +1, -1, -1, -1, -1, +1, -1}				+1 = mark, -1 = space
F_M	Mark Frequency	131.236875	131.25	131.263125	kHz	$6.25\text{kHz} \times 21$
F_S	Space Frequency	143.735625	143.75	143.764375	kHz	$6.25\text{kHz} \times 23$
T_S	Average Bit Period	5.119488	5.12	5.120512	ms	(Time to complete one full duty cycle)/219
T_T	Transmission Period	168.943104	168.96	168.976896	ms	3 Words
T_Q	Quiet Period	901.029888	901.12	901.210112	ms	16 Words
T_C	Cycle Period	1069.972992	1070.08	1070.187008	ms	19 Words



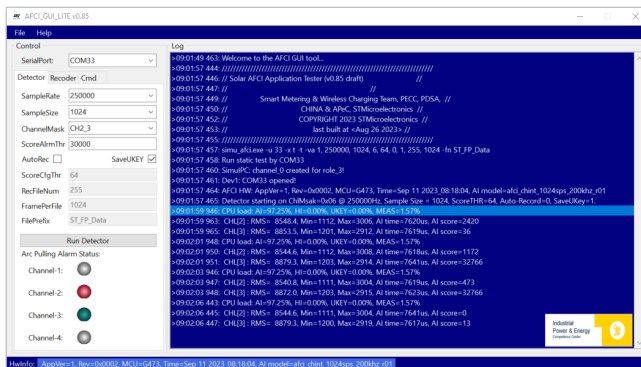
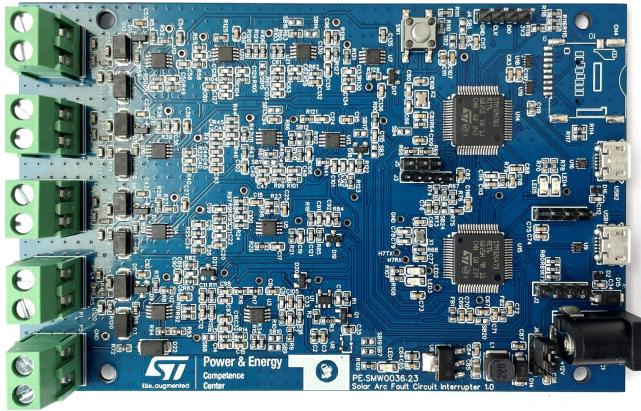
AFCI数据处理潜在风险



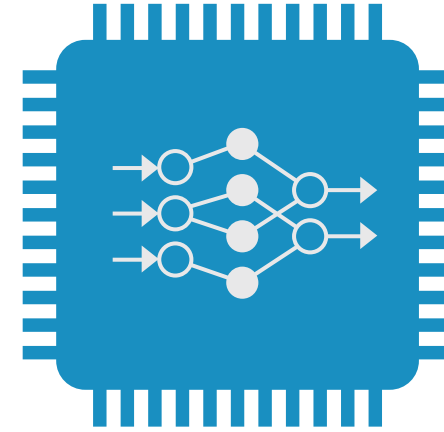
现场数据采集问题示例

ST电弧故障分断器参考设计

Turn-key HW / SW Reference Design



Turn-key STM32 Embedded AI



NANOEDGE AI
STUDIO



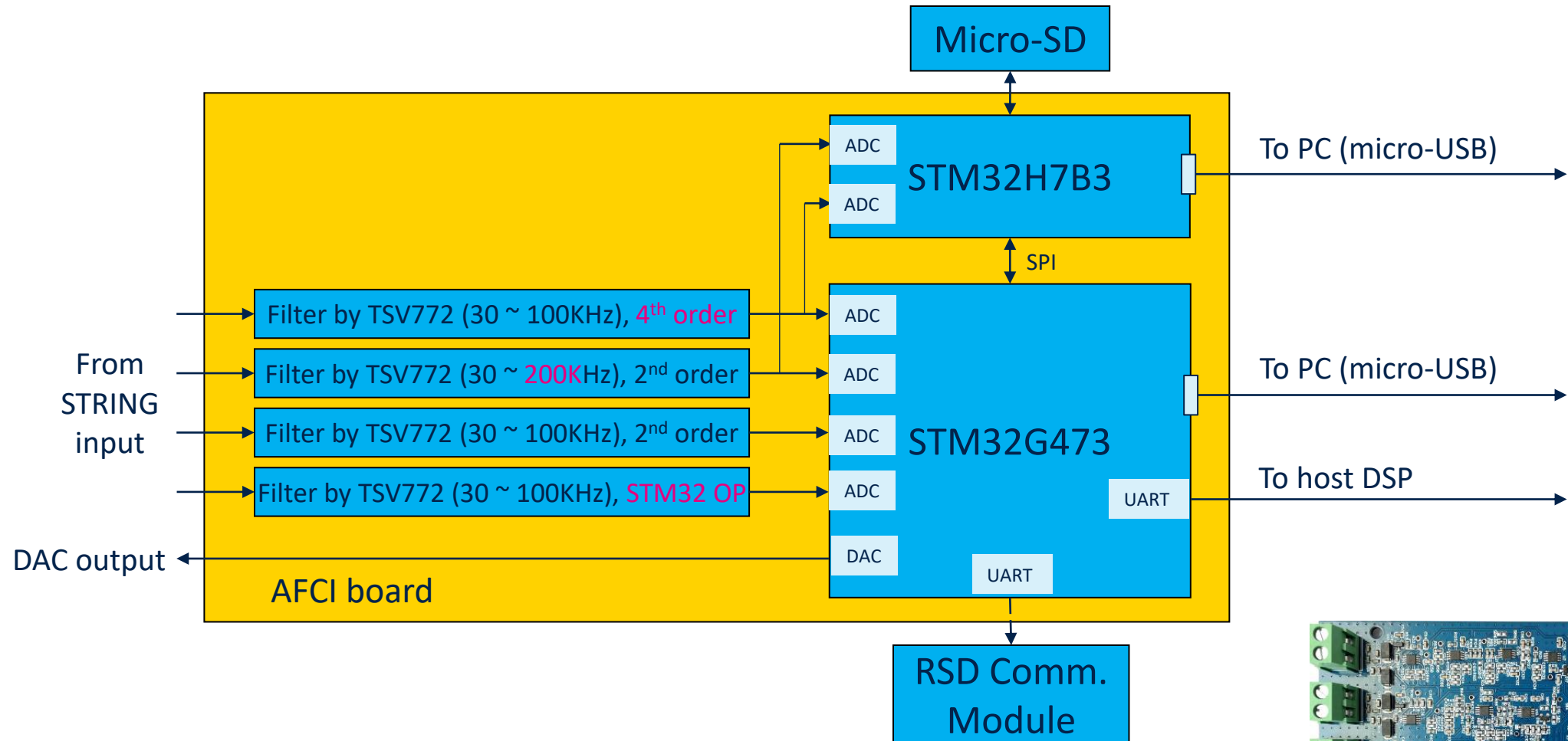
STM32
Cube.AI



All STM32
MCUs

AFCI硬件与固件的参考设计

AFCI电路板硬件框图

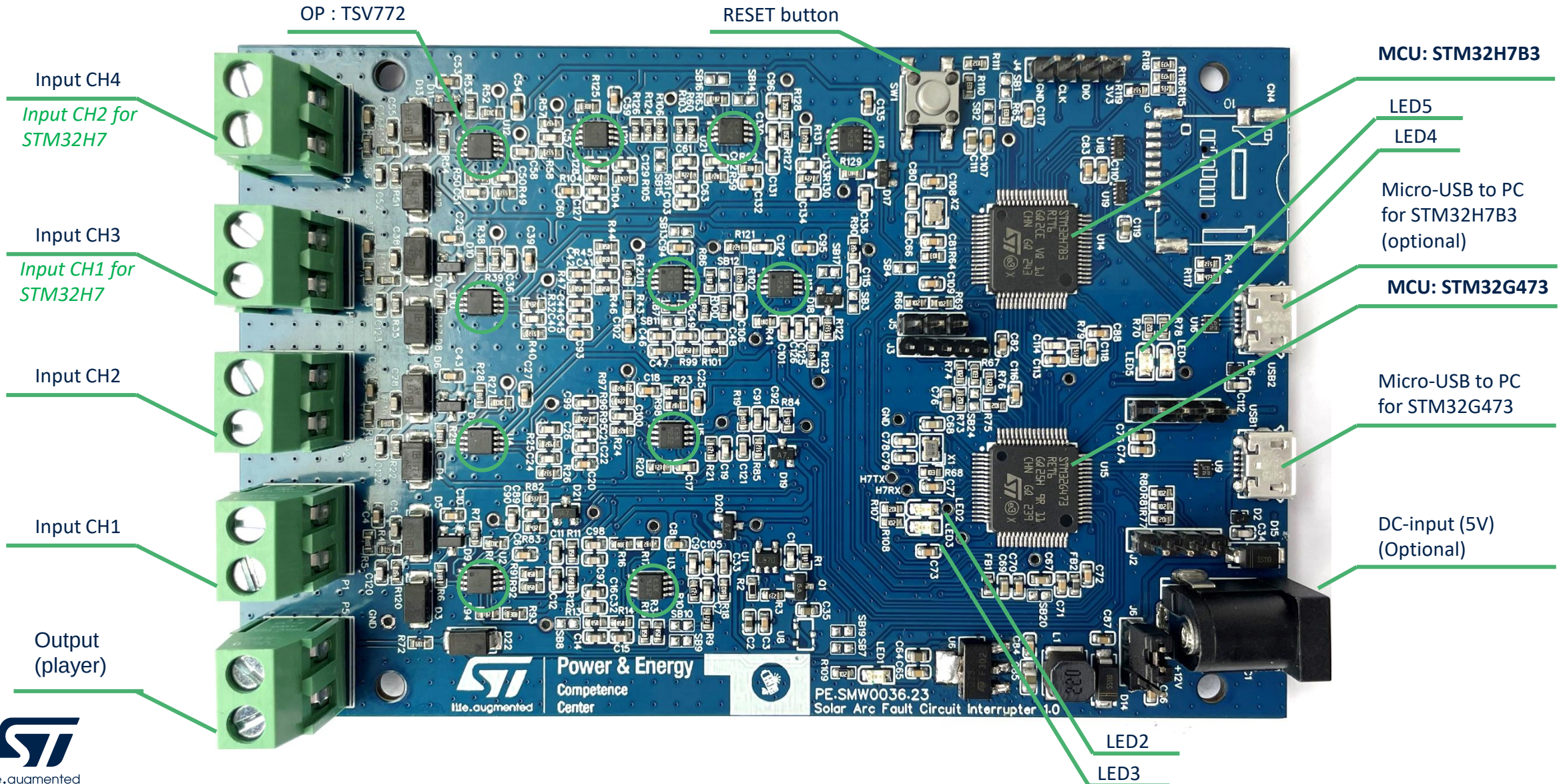


STM32H7B3主要用于驱动micro-SD卡数据存储。

STM32G473或STM32H7B3任选其一都足以实现客户AFCI产品。

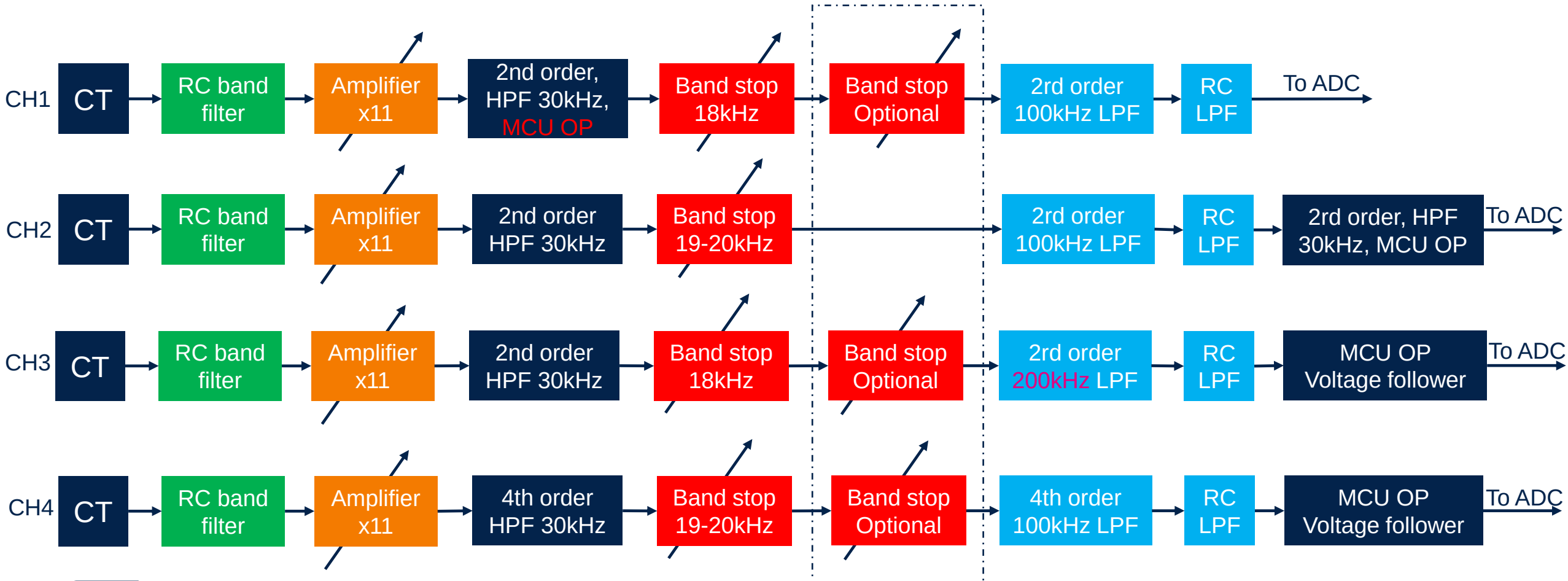


AFCI硬件电路板v1.0

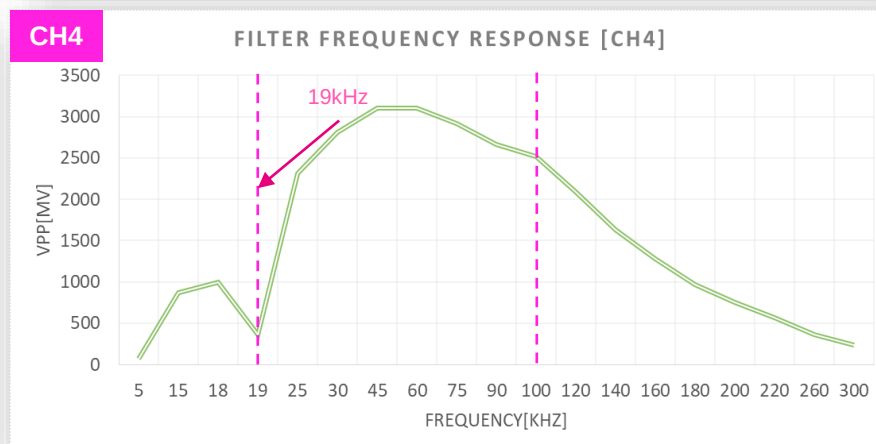
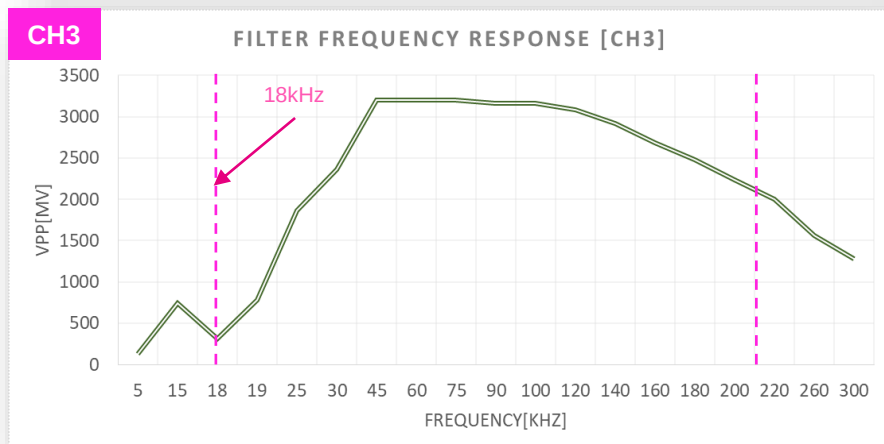
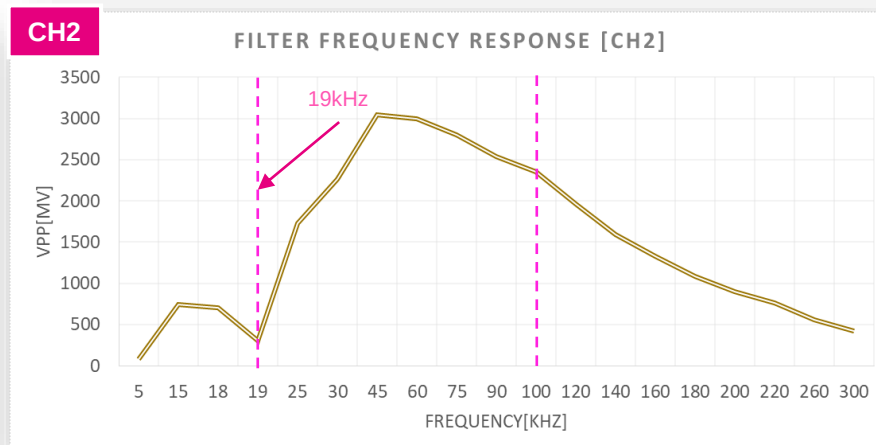
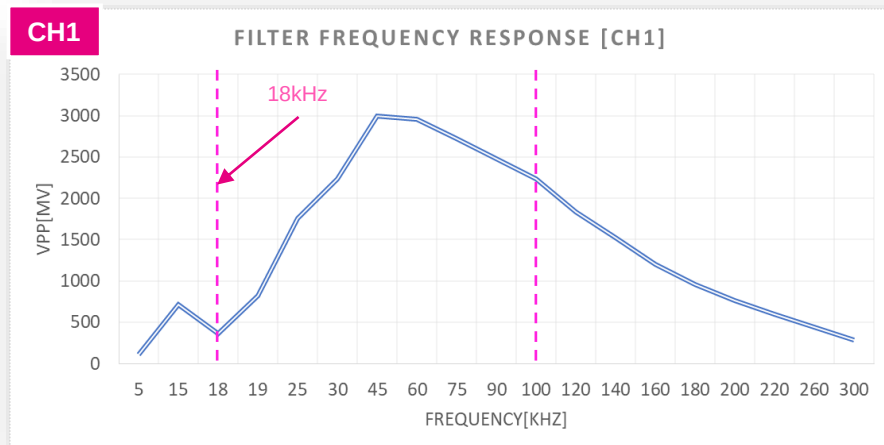


AFCI滤波器设计

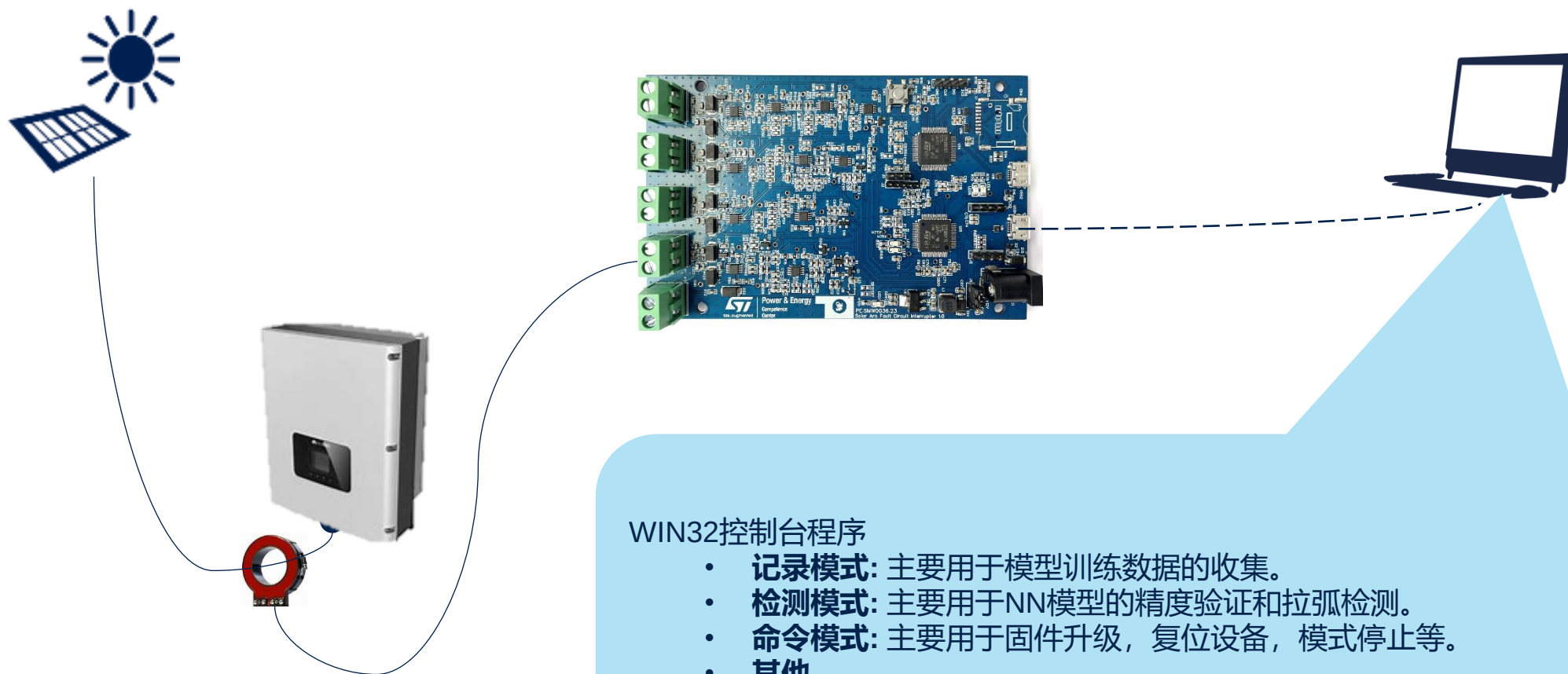
当前电路板中的第二级带阻滤波器被跳线帽旁路



硬件滤波器的频率响应



系统连接



WIN32控制台程序

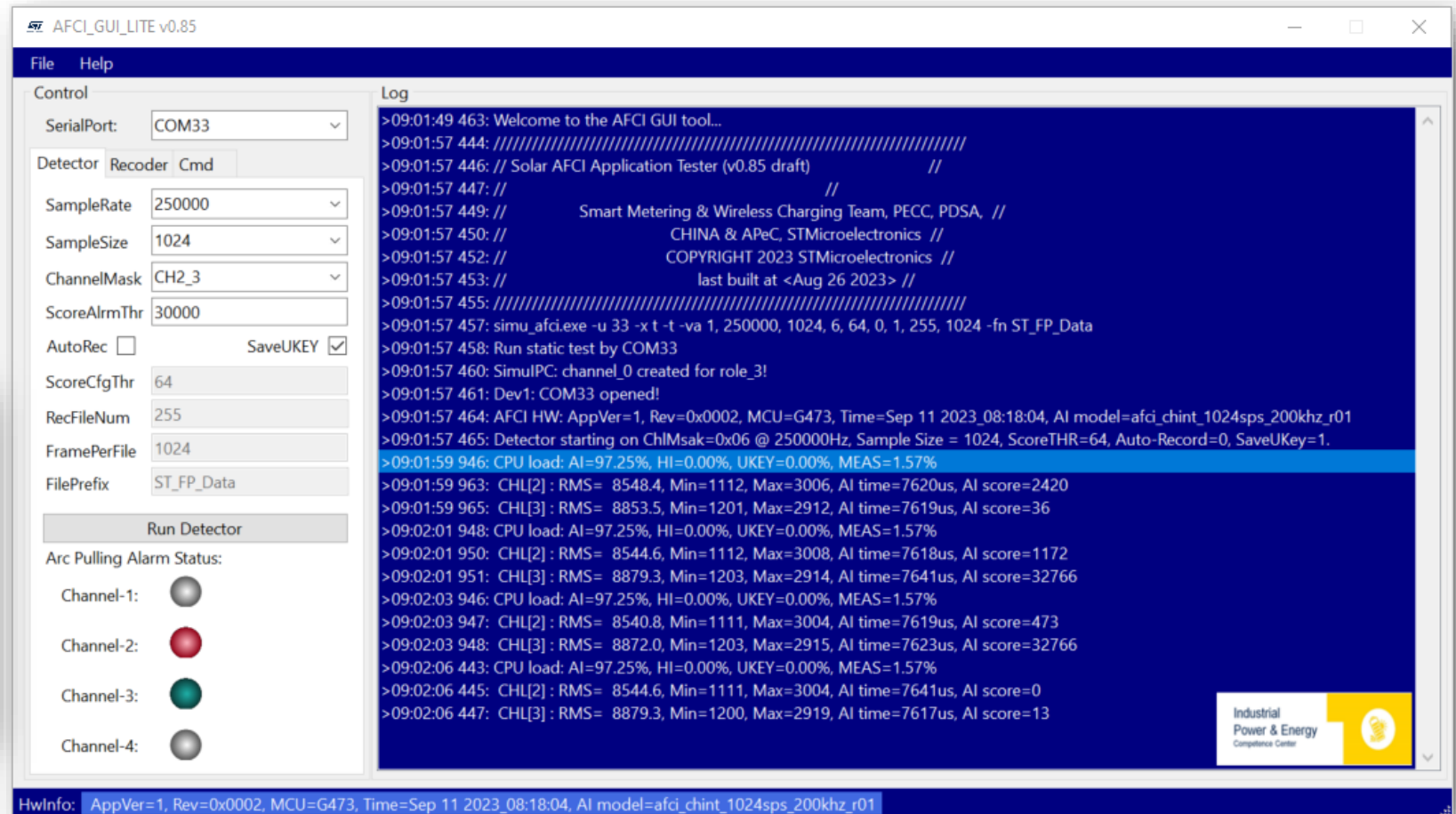
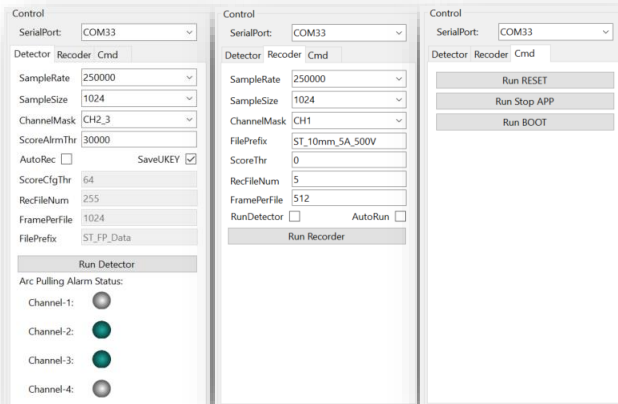
- **记录模式:** 主要用于模型训练数据的收集。
- **检测模式:** 主要用于NN模型的精度验证和拉弧检测。
- **命令模式:** 主要用于固件升级, 复位设备, 模式停止等。
- **其他 ...**

AFCI图形化用户接口工具

AFCI 图形化用户接口

AFCI 图形化用户接口工具让开发更简单

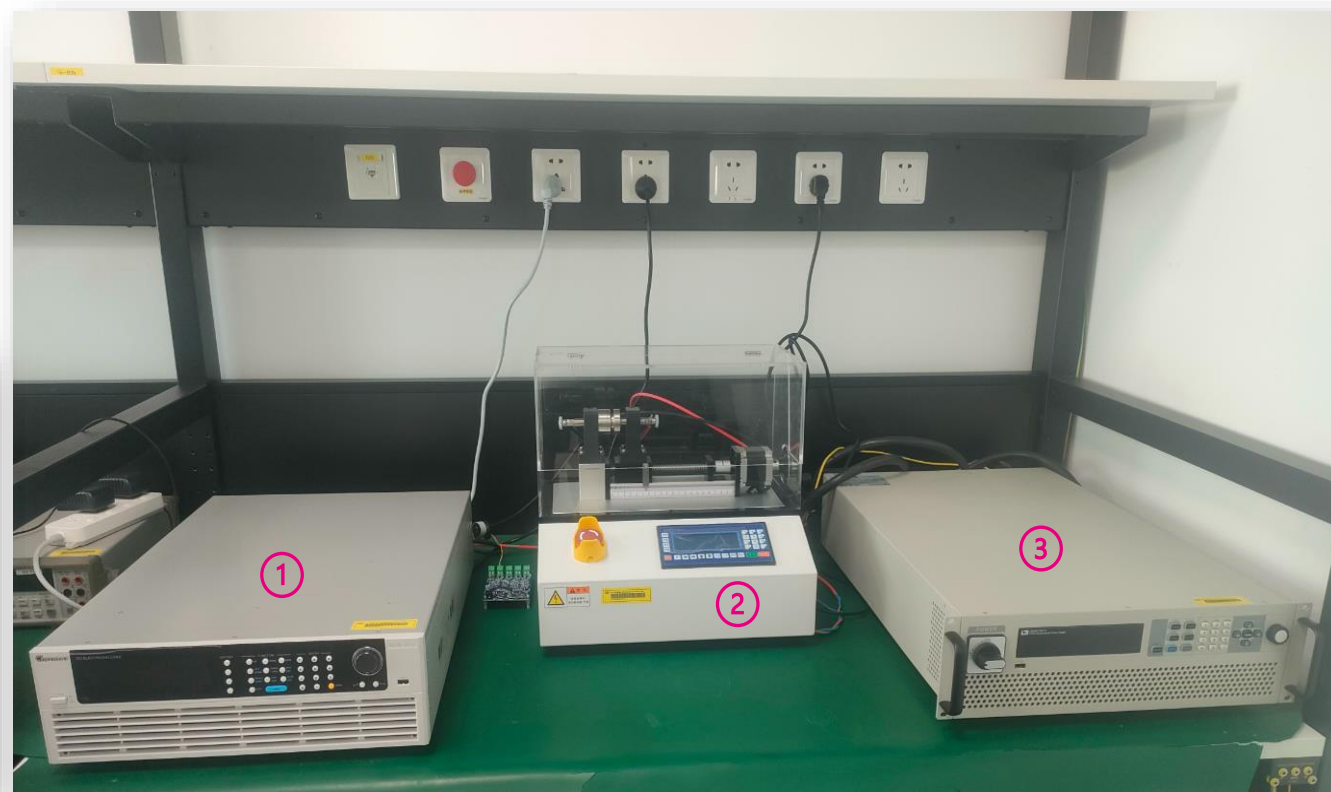
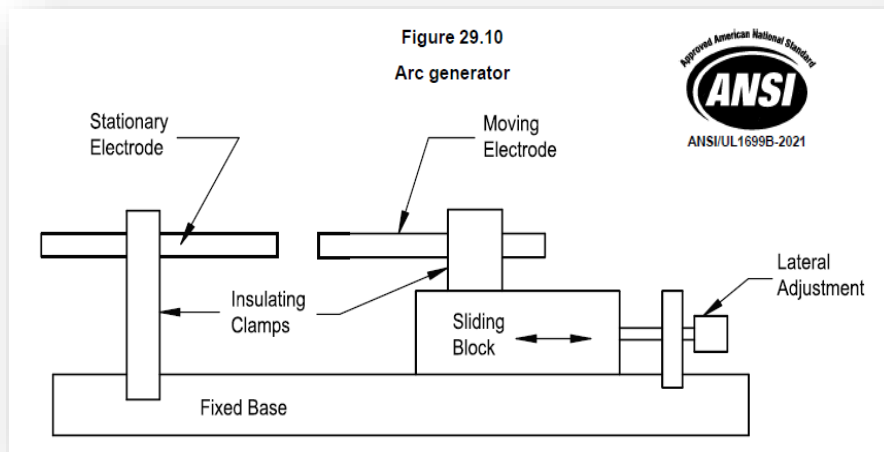
- 支持检测模式
- 支持采集模式
- 支持控制命令
- 支持NVM配置
- 支持日志存储
-



ST 电弧故障检测实验室

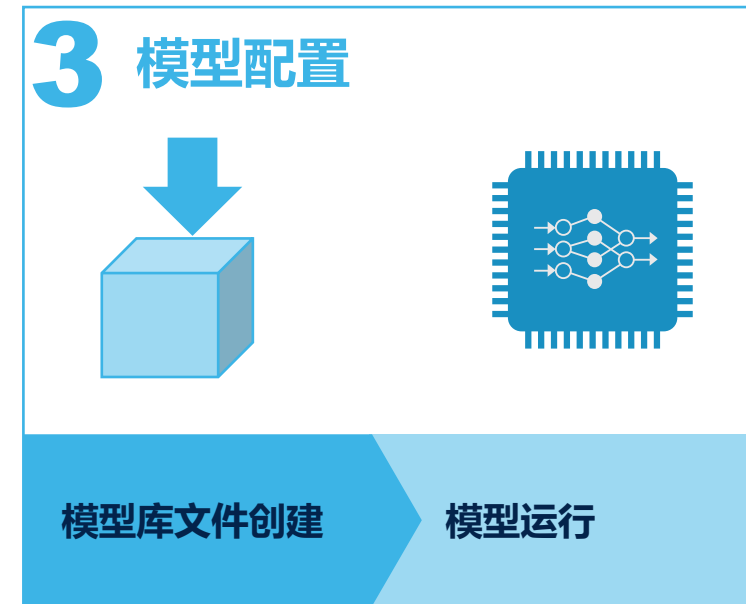
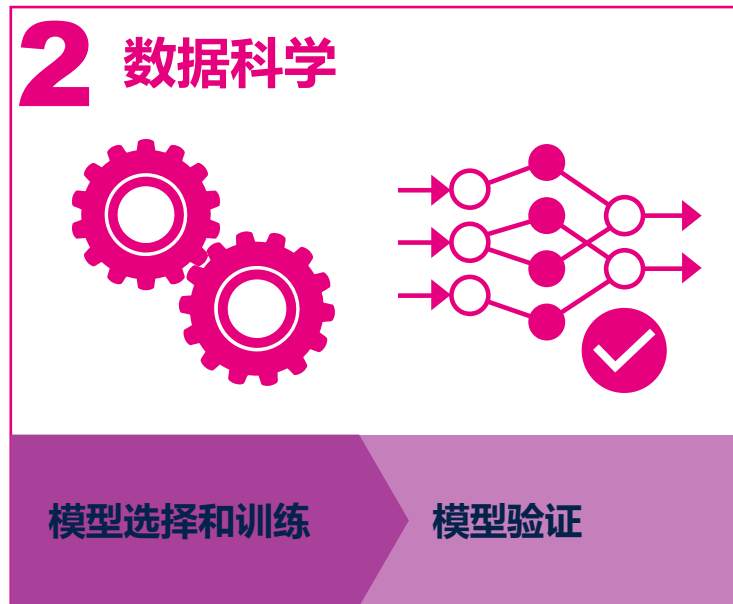
AFCI实验室的主要仪器列表:

1. 直流电子负载
2. 电弧发生器
3. 直流供电电源



AI 参考设计

整体的AI开发流程



NANOEDGE AI
STUDIO

自动化的边缘AI软件

STM32
Cube.AI

STM32
Cube.AI

边缘 AI 工具包



All STM32
MCUs

AFCI 项目流程



ST提供的AFCI开发包

AFCI硬件

参考板1.0

包含STM32G4 和 STM32H7的硬件电路板用于数据收集和测试



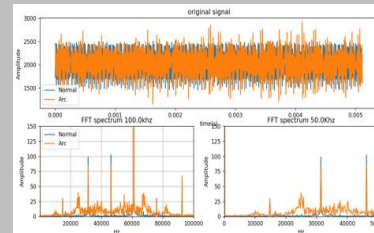
IN32控制台应用程序

- 数据收集程序
- 实时检测程序
- 软件更新的程序

```
run_send_cmd.bat
run_test_detector.bat
run_test_record.bat
simu_afci.exe
```

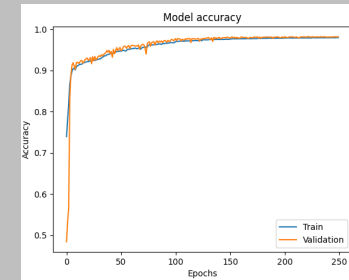
信号检测工具 (exe 文件)

用于分析采集到数据的信号质量



AFCI AI 模型 训练

(exe 文件)
用于训练采集的数据来生成AI模型



针对STM32的 AFCI 软件项目包

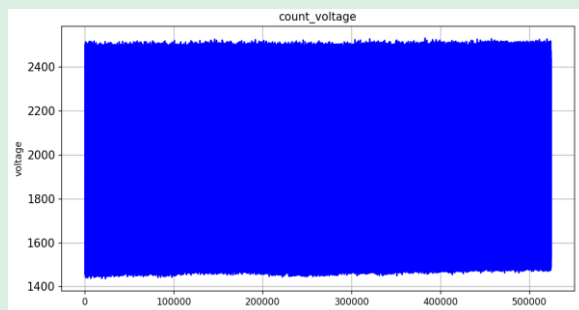
- 数据记录工具
- 模型实时处理方法
- 相关的后处理

❖ 用户手册及指引

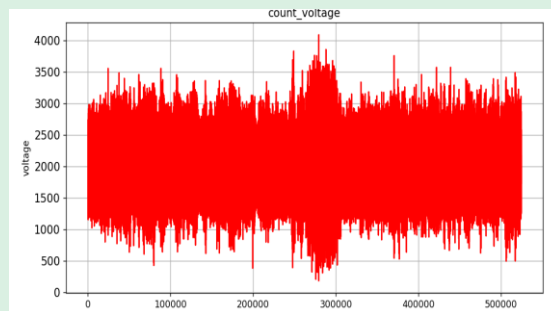
信号检查

在训练神经网络之前，使用信号检测工具检查信号是非常重要的。

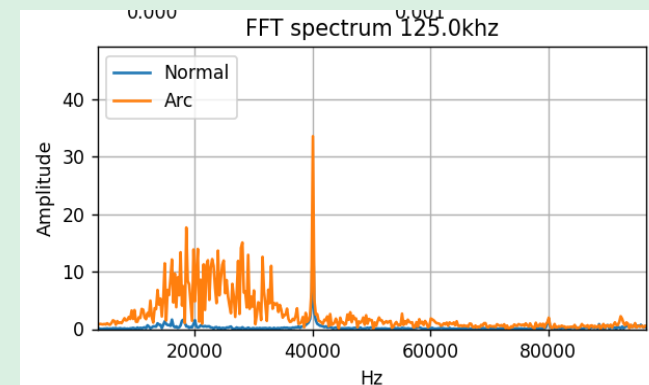
正常信号示例



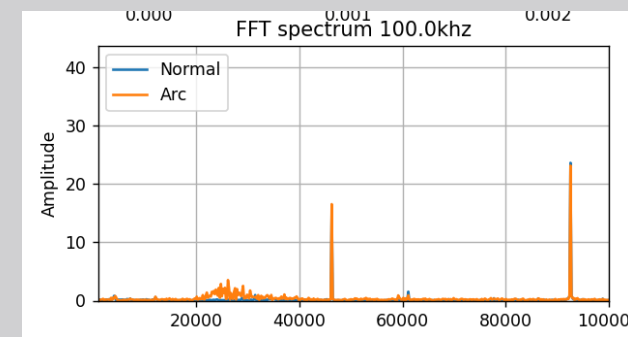
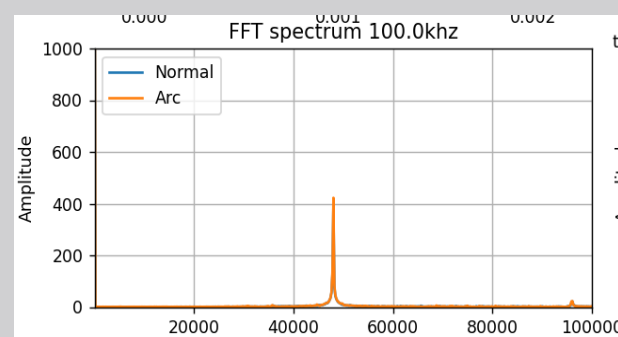
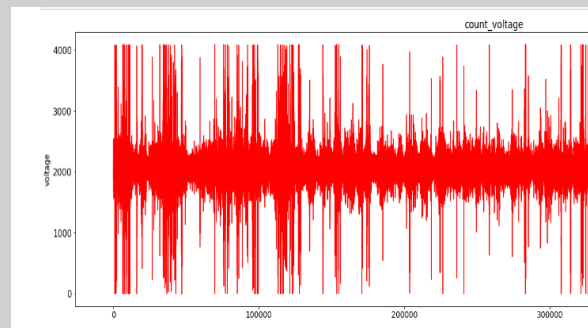
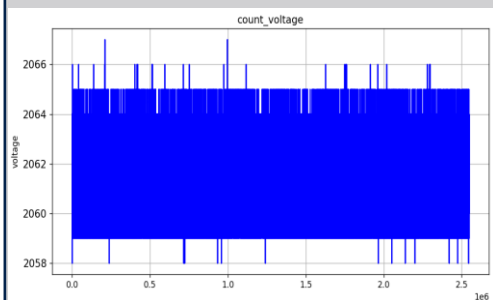
Normal signal



arc signal



异常信号示例



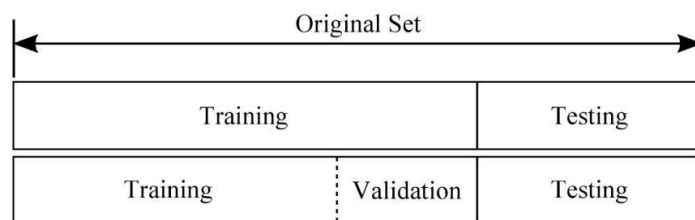
AI 模型训练

- 使用AFCI GUI的工具来收集**两类数据集**

- 正常状态信号
- 拉弧异常信号

- 数据集分割

- 训练集
- 验证集
- 测试集



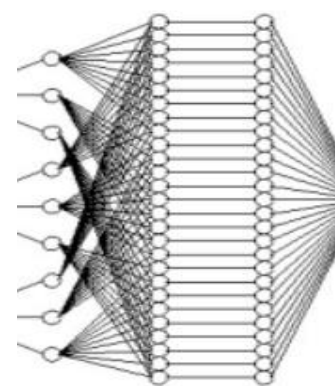
打标数据集

训练集

验证集

测试集

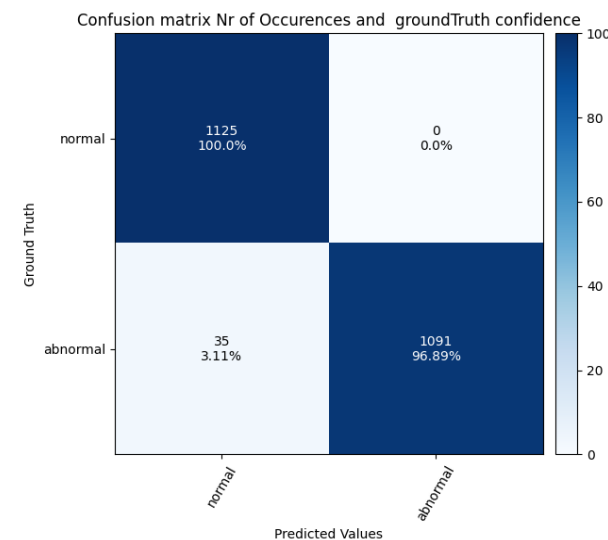
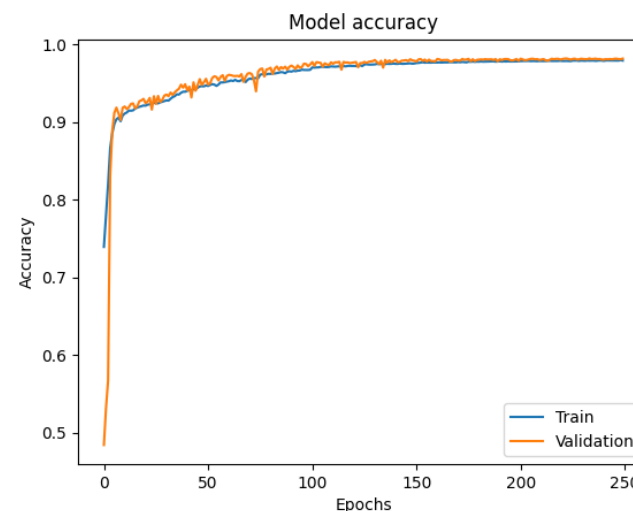
神经网络结构



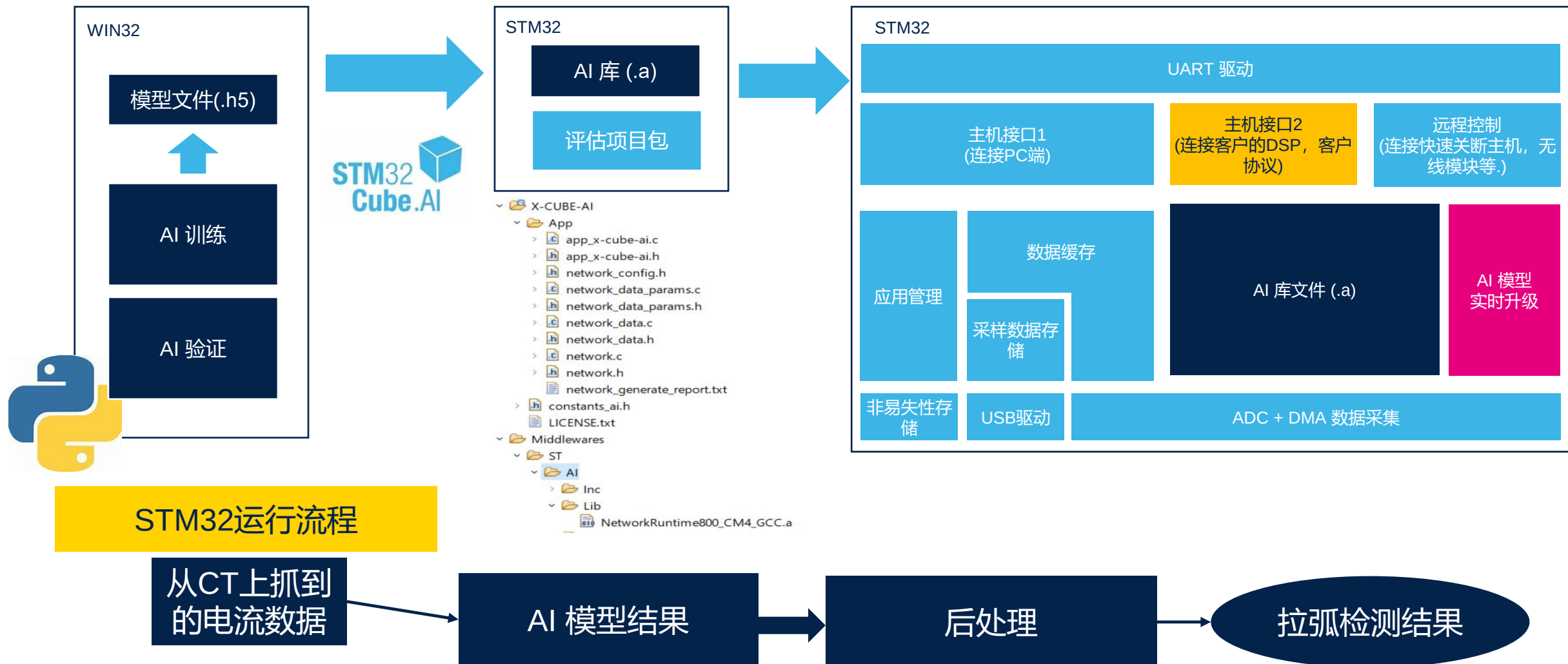
Softmax

分类1
(正常信号)

分类2
(拉弧异常信号)



AI 模型实施



基于AI的电弧故障断路器 (AFCI) 助力于安全和可持续发展。

Turn-Key 参考设计方案，能有效快速部署最先进的AFCI方案

基于深度学习的检测器，可达到高检测准确率

高性能基于边缘计算的STM32G4 或是 STM32H7的MCU

TSV7xx 运放带来更灵活和准确的信号处理

经验丰富的支持团队，提供成熟的材料和流程



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